AI2200: CONCENTRATION INEQUALITIES

PRACTICE PROBLEMS

భారతీయ సాంకేతిక విజ్ఞాన సంస్థ హైదరాబాద్
भारतीय प्रौद्योगिकी संस्थान हैदराबाद
Indian Institute of Technology Hyderabad

TOPICS: Bernoulli Concentrations, Basic Probability

1. (a) Suppose that $X, Y \overset{\text{IID}}{\sim}$ Exponential(1). Let $Z = \max\{X, Y\}$. What is the distribution of Z?

   (b) Fix $n \in \mathbb{N}$ and $\lambda > 0$, and suppose that $X_1, \ldots, X_n \overset{\text{IID}}{\sim}$ Exponential(λ).
   What is the distribution of $Z_n = \max\{Z_1, \ldots, Z_n\}$?

   (c) In (a), the maximum of X and Y is one of either X or Y. Yet, while both X and Y have Exponential(1) distribution, the distribution of Z is not Exponential(1). How do you explain this discrepancy?

   (d) In (b), the maximum of $X_1, \ldots, X_n$ is one of these random variables. Yet, while each of X_1, X_2, etc have Exponential(1) distribution, the distribution of Z_n is not Exponential(1). How do you explain this discrepancy?

   (e) For the setting in part (a), show that for any $\varepsilon > 0$,
   $$\mathbb{P}(\{X \geq \varepsilon\}) \;=\; \mathbb{P}(\{Y \geq \varepsilon\}) \;\leq\; \exp(-\varepsilon).$$
   Furthermore, show that for any $\varepsilon > 0$,
   $$\mathbb{P}(\{Z \geq \varepsilon\}) \;\not\leq\; \exp(-\varepsilon), \qquad \mathbb{P}(\{Z \geq \varepsilon\}) \;\leq\; 2\exp(-\varepsilon).$$

   (f) For the setting in part (b), show that for any $\varepsilon > 0$,
   $$\mathbb{P}(\{X_1 \geq \varepsilon\}) = \cdots = \mathbb{P}(\{X_n \geq \varepsilon\}) \;\leq\; \exp(-\varepsilon),$$
   whereas
   $$\mathbb{P}(\{Z_n \geq \varepsilon\}) \;\not\leq\; \exp(-\varepsilon), \qquad \mathbb{P}(\{Z_n \geq \varepsilon\}) \;\leq\; n\exp(-\varepsilon).$$

   Note: The point of part (e) above is that although exponential upper bound of $\exp(-\varepsilon)$ holds for each of X and Y, the same exponential upper bound does not hold for a function of X, Y such as Z. However, a larger exponential upper bound may be obtained for Z. A similar explanation holds for Z_n in part (f) which is a function of the IID data $X_1, \ldots, X_n$.

2. (a) Show that for all $\delta \geq 0$,
   $$\log(1+\delta) \;\geq\; \frac{2\delta}{2+\delta}.$$
   Hint: Show that the function $f(\delta) = \log(1+\delta) - \dfrac{2\delta}{2+\delta} \geq 0$ for all $\delta \geq 0$.

   (b) Along similar lines as part (a), show that for all $\delta \in [0, 1]$,
   $$\delta + (1-\delta)\,\log(1-\delta) \;\geq\; \frac{\delta^2}{2}.$$

   (c) Fix $p \in (0, 1)$ and $\delta > 0$ such that
   $$0 \;\leq\; p(1+\delta) \;\leq\; 1.$$
   Consider the Kullback–Leibler divergence
   $$D_{\mathsf{KL}}(p(1+\delta)\|p) \;=\; p(1+\delta)\,\log(1+\delta) + (1-p-p\delta)\,\log\left(\frac{1-p-p\delta}{1-p}\right).$$
   Using the result of part (a) and the inequality $\log x \geq 1 - \frac{1}{x}$ for all $x > 0$, show that
   $$D_{\mathsf{KL}}(\mathrm{Ber}(p(1+\delta))\|\mathrm{Ber}(p)) \;\geq\; \frac{p\delta^2}{2+\delta}.$$

(d) Fix $p \in (0,1)$ and $\delta > 0$ such that

$$0 \;\leq\; p(1-\delta) \;\leq 1.$$

Consider the Kullback–Leibler divergence

$$D_{\mathsf{KL}}(\mathsf{Ber}(1-p(1-\delta)) \,\|\, \mathsf{Ber}(1-p)) \;=\; (1-p(1-\delta))\ \log\left(\frac{1-p(1-\delta)}{1-p}\right) + p(1-\delta)\ \log(1-\delta).$$

Using the result of part (b) and the inequality $\log x \geq 1 - \frac{1}{x}$ for all $x > 0$, conclude that

$$D_{\mathsf{KL}}(\mathsf{Ber}(1-p(1-\delta)) \,\|\, \mathsf{Ber}(1-p)) \;\geq \frac{p\delta^2}{2} \;\geq\; \frac{p\delta^2}{2+\delta}.$$

(e) Combining the results of parts (c) and (d), conclude that if $X_1, \ldots, X_n \overset{\text{IID}}{\sim} \mathsf{Ber}(p)$, then

$$\mathbb{P}\left(\left\{\left|\frac{S_n}{n} - p\right| > \delta\right\}\right) \;\leq\; 2\ \exp\left(-\frac{np\delta^2}{2+\delta}\right) \qquad \forall\, \delta > 0.$$

3. Let $\mathcal{X}$ be an arbitrary feature space, and let $\mathcal{Y}$ be a finite label space.
Let $P_{X,Y}$ be a probability distribution on $\mathcal{X} \times \mathcal{Y}$.
Let $\mathcal{G} = \{g_1, \ldots, g_M\}$ be a finite collection of classifiers mapping $\mathcal{X}$ to $\mathcal{Y}$.
Let $(X_1, Y_1), \ldots, (X_n, Y_n) \overset{\text{IID}}{\sim} P_{X,Y}$, and for each $g \in \mathcal{G}$, let

$$R(g) \;\coloneqq\; \mathbb{P}_{(X,Y)\sim P_{X,Y}}\ \mathbb{P}\Big(\big\{g(X) \neq Y\big\}\Big), \qquad\qquad \widehat{R}_n(g) \;\coloneqq\; \frac{1}{n}\sum_{k=1}^{n} \mathbf{1}_{\{g(X_k)\neq Y_k\}}.$$

Let $g^\star$ and $\widehat{g}_n$ be defined as

$$g^\star \;\coloneqq\; \operatorname*{arg\,min}_{g\in\mathcal{G}}\ R(g), \qquad\qquad \widehat{g}_n \;\coloneqq\; \operatorname*{arg\,min}_{g\in\mathcal{G}}\ \widehat{R}_n(g).$$

(a) Show formally that for each $g \in \mathcal{G}$ and for each $\varepsilon > 0$,

$$\mathbb{P}\left(\left\{\left|\widehat{R}_n(g) - R(g)\right| > \varepsilon\right\}\right) \;=\; \mathbb{P}\left(\left\{\left|\frac{1}{n}\sum_{k=1}^{n} \mathbf{1}_{\{g(X_k)\neq Y_k\}} - R(g)\right| > \varepsilon\right\}\right) \;\leq\; 2\ \exp\left(-\frac{n\varepsilon^2 R^2(g^\star)}{2+\varepsilon}\right).$$

(b) Although $\widehat{g}_n$ is one among $\{g_1, \ldots, g_M\}$, argue why the result of part (a) does not hold with $g = \widehat{g}_n$.
Can you see the connection of this question with question 1(f), noting that $\widehat{g}_n$ is a function of $X_1, \ldots, X_n$.

(c) In class, we saw (via the union bound) that

$$\mathbb{P}\left(\left\{\left|\widehat{R}_n(\widehat{g}_n) - R(\widehat{g}_n)\right| > \varepsilon\right\}\right) \;\leq\; \sum_{g\in\mathcal{G}} \mathbb{P}\left(\left\{\left|\widehat{R}_n(g) - R(g)\right| > \varepsilon\right\}\right) \;\leq\; 2|\mathcal{G}|\ \exp\left(-\frac{n\varepsilon^2 R^2(g^\star)}{2+\varepsilon}\right).$$

Explain clearly why $\widehat{g}_n$ is not one of the elements in the summation of the middle term above. If $\widehat{g}_n$ were indeed one of the elements of the summation, then the summation would still be a random quantity (because $\widehat{g}_n$ is random) while the left-hand side is a deterministic number.

4. Consider a coin having an unknown bias p.
Suppose that only one of the following hypotheses is true about the unknown bias p:

$$\mathcal{H}_0:\ \ p = 0.25, \qquad\qquad \mathcal{H}_1:\ \ p = 0.5.$$

We wish to identify the true hypothesis by tossing the coin multiple times.
Let $X_1, \ldots, X_n \overset{\text{IID}}{\sim} \mathsf{Ber}(p)$ denote the outcomes resulting from tossing the coin n times.

(a) Assuming $\mathcal{H}_0$ were true, evaluate the likelihood

$$\mathbb{P}_{\mathcal{H}_0}(X_1, \ldots, X_n).$$

Express your answer in terms of $X_1, \ldots, X_n$.

(b) Assuming $\mathcal{H}_1$ were true, evaluate the likelihood

$$\mathbb{P}_{\mathcal{H}_1}(X_1, \ldots, X_n).$$

Express your answer in terms of $X_1, \ldots, X_n$.

(c) Let $T : \{0,1\}^n \to \{\mathcal{H}_0, \mathcal{H}_1\}$ be defined as

$$T(X_1, \ldots, X_n) \;=\; \begin{cases} \mathcal{H}_1, & \log \dfrac{\mathbb{P}_{\mathcal{H}_1}(X_1, \ldots, X_n)}{\mathbb{P}_{\mathcal{H}_0}(X_1, \ldots, X_n)} \;\geq\; 1, \\ \\ \mathcal{H}_0, & \text{otherwise.} \end{cases}$$

Furthermore, let

$$R_n \;=\; \max\Big\{\mathbb{P}_{\mathcal{H}_0}\Big(\big\{T(X_1, \ldots, X_n) = \mathcal{H}_1\big\}\Big),\; \mathbb{P}_{\mathcal{H}_1}\Big(\big\{T(X_1, \ldots, X_n) = \mathcal{H}_0\big\}\Big)\Big\}.$$

i. Using Bernoulli concentration result of question 2, derive an exponential upper bound on each of the probability terms

$$\mathbb{P}_{\mathcal{H}_0}\Big(\big\{T(X_1, \ldots, X_n) = \mathcal{H}_1\big\}\Big), \qquad\qquad \mathbb{P}_{\mathcal{H}_1}\Big(\big\{T(X_1, \ldots, X_n) = \mathcal{H}_0\big\}\Big).$$

ii. Given $\delta \in (0,1)$, use your result in part (i) above to compute the minimum number of samples n (in terms of δ) required to guarantee $R_n \leq \delta$.